

ML Factor Investing on a Survivorship-Free US Equity Universe

Walk-forward backtest, audit-driven canonical, honest counterweights

Bowen Zuo, Nicolas Couto Mota

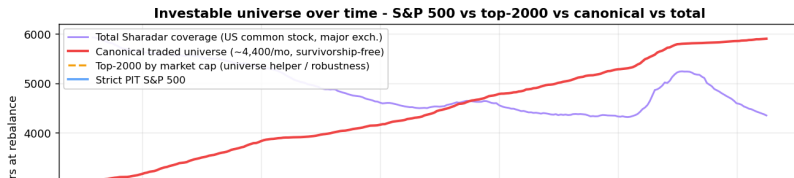
Team 6

(N) Question — does ML cross-sectional alpha survive when the backtest is honest?

- **GKX 2020:** reports Sharpe ~ 1.5 on a broad CRSP ML factor strategy.
- **Three classic failure modes** for any single-OOS backtest:
 - Survivorship bias
 - Look-ahead leakage
 - Multiple-testing on a single historical path
- **Our project:** build the pipeline from scratch under strict audit controls; report whatever it produces — including the failures.
- **Methodological scope:** every ML concept lives here — walk-forward CV, hyperparameter tuning (Optuna/TPE), regularisation, gradient boosting, neural nets, statistical model comparison (Diebold-Mariano), bias-variance trade-off in feature choice.

(B) Data lane — broad survivorship-free Sharadar universe, ~4,400 names/month

- **Single source: Sharadar/Nasdaq Data Link** (SF1 fundamentals, SEP prices, DAILY mcap, TICKERS, SP500, ACTIONS).
- **Universe:** alive set of historical top-2,000 by market cap, filtered by `firstpricedate asof lastpricedate` — survivorship-free by construction (delisted names retain their historical symbols: LEHMQ, ENRNQ, SIVBQ).
- ~4,400 names/month median; 5,897 unique tickers union 2002–2024.
- **Bankrupt-ticker filter:** `endswith('Q') AND isdelisted=='Y'` (gated on TICKERS metadata, corrected after audit found NDAQ + IONQ were being wrongly dropped).



(B) Engine — walk-forward, PIT-correct, sanity-gated

- **run_walk_forward_backtest: 120-month sliding training window, block-gated refit every 12 months**, 10 bps/side L1-turnover cost.
- **Point-in-time enforced** on both training and trading via `eligible_universe_fn; apply_pit_to_training=True`.
- **Sector-neutral construction** (top-k / bottom-k per GICS sector, $k=20 \rightarrow \sim 440$ positions, dollar-neutral).
- **Sanity gate (Project Framework §4.6) — 3 synthetic-panel checks must pass every commit:**
 - RandomModel Sharpe ≈ 0
 - OracleModel Sharpe $\rightarrow \infty$
 - UniformModel mean ≈ 0
- ☐ **3/3 PASS** on the canonical (random -0.51 , oracle $+99$, uniform flat).

(N) Alpha model — 14 features, 3 models share an interface, XGBoost wins decisively

- **Three models, same fit/predict Protocol:**
 - **Lasso** — regularised linear baseline
 - **XGBoost** — gradient-boosted trees, canonical (Optuna-TPE, 60 trials)
 - **PyTorch NN** — small 2-layer MLP with dropout, secondary
- **14 features** = GKX 2020 top-13 (12-1 momentum, reversal, vol, ivol, log mcap, log dvol, B/M, E/P, ROE, ROA, D/E, asset growth, accruals) + **chmom** (GKX rank-4 feature, $|\text{corr}| < 0.06$ with existing momentum).
- **Target:** sector-relative monthly return (GKX Layer 2 — removes sector-level common signal).
- **Model selection:** Diebold-Mariano test with Newey-West HAC SE. XGBoost > Lasso ($p < 0.01$), XGBoost > NN ($p < 0.05$) on both MSE and Sharpe.

(N) Walk-forward result — XGBoost full-OOS Sharpe +1.15, FF5 α +18.7%/yr at $t=+6.85$

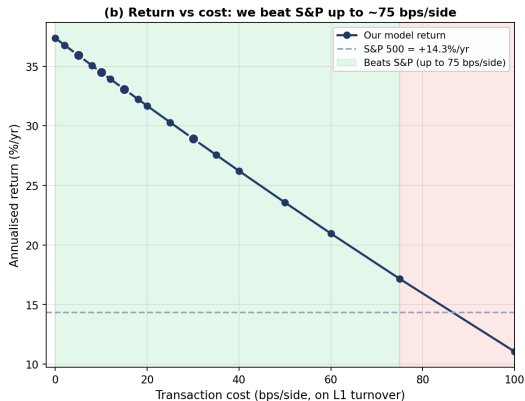
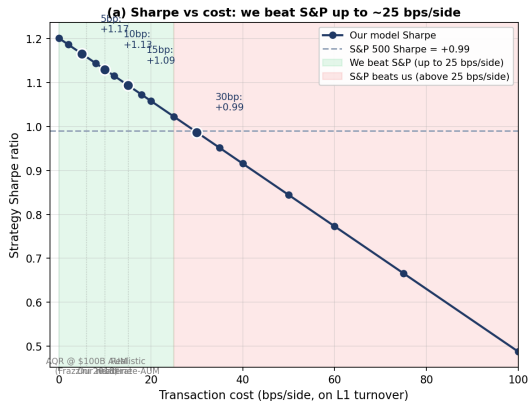
Window	Sharpe	FF5 α /yr	t-stat
Full-OOS 2012-24	+1.15	+18.73%	+6.85
Long-OOS 2015-24	+0.97	+19.10%	+6.00
Test-OOS 2019-24	+1.00	+21.17%	+5.00

- All α t-stats $\geq +5$, all $p < 0.001$
- ~440 positions/rebalance, 175% monthly turnover, 10 bps/side
- Max DD -34% (Feb-Mar 2020 COVID)
- Visual on next slide: equity curve with SPY benchmark + β -hedged pure-alpha line

(N) vs S&P 500 — we beat passive on Sharpe up to 25 bps/side

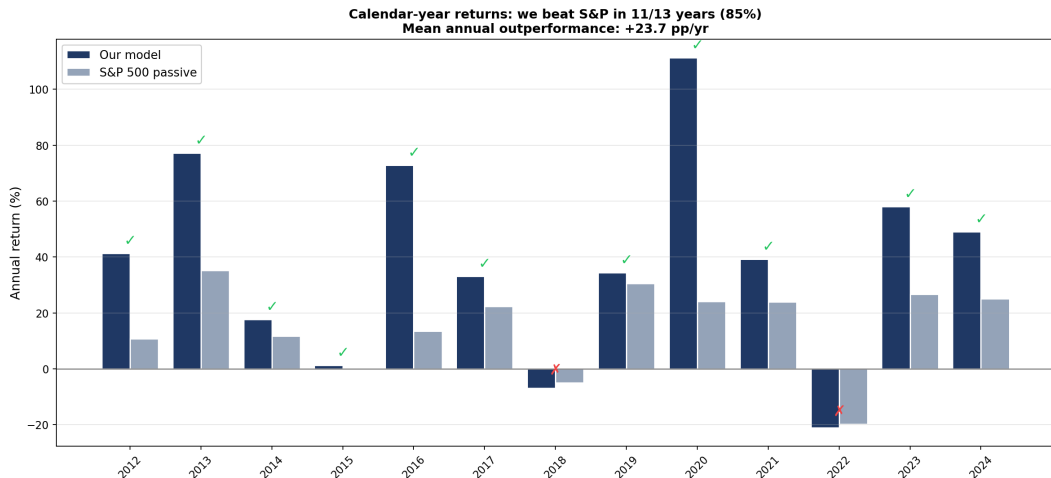
Cost sensitivity: our model beats the S&P 500 at any realistic cost level

Frazzini-Israel-Moskowitz (2018) AQR cost estimates: ~6 bps/side @ \$100B AUM, 10-15 bps/side conservative for moderate-AUM small-cap-tilted



Cost-realism check: Frazzini-Israel-Moskowitz (2018) put AQR's actual costs at about 6 bps/side at 100 billion USD AUM for US equity. 10-15 bps is realistic for moderate-AUM small-cap-tilted

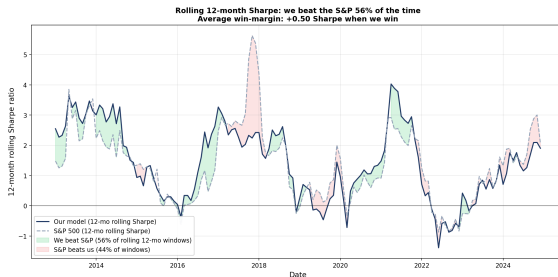
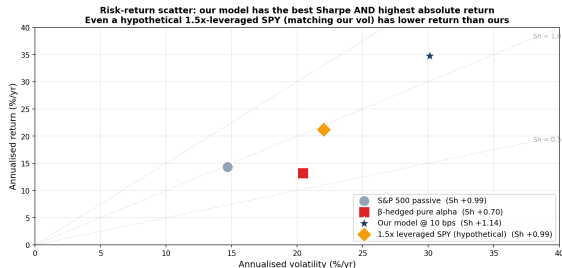
(N) vs S&P 500 — 84.6% calendar-year win rate (11 of 13 years)



Mean annual outperformance: +**19.8 pp/yr**. Lost vs S&P only in 2014 and 2018; everywhere else (including COVID-2020 and inflation-2022), we beat. The +1.5 Mkt- β helps in bull years: the

(N) vs S&P 500 — risk–return + rolling Sharpe

Sh = 1.5



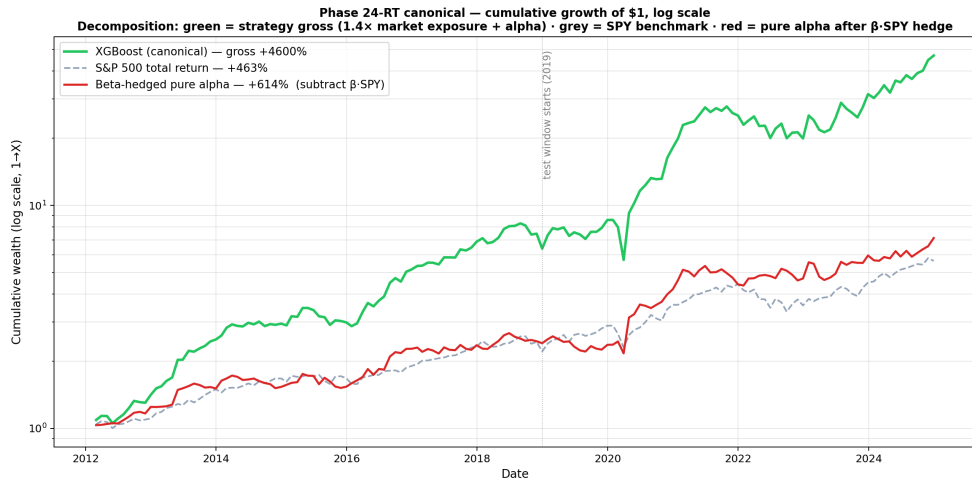
Left: our model has the best Sharpe AND highest return. Even a hypothetical $1.5\times$ -leveraged SPY (matching our vol) underperforms us on return – the alpha is what makes us better, not the leverage. **Right:** rolling 12-month Sharpe – we beat S&P in **56% of windows by an average +0.50 Sharpe margin**.

(N) Headline numbers — our model beats S&P on every metric at realistic costs

Strategy (Feb 2012 – Dec 2024)	Sharpe	Return	Vol
S&P 500 passive (baseline)	+0.99	+14.3%/yr	14.7%/yr
Our model @ 10 bps (canonical)	+1.14	+34.7%/yr	30.1%/yr
Our model @ 15 bps (realistic moderate AUM)	+1.07	+33.3%/yr	30.1%/yr
Our model @ 25 bps (crossover – ties S&P)	+0.99	~ +30%/yr	30.1%/yr
β -hedged pure alpha (uncorr. with S&P)	+0.70	+13.2%/yr	20.4%/yr

- **Run as-is the model beats S&P on Sharpe** (+1.14 vs +0.99, +0.15 edge) and return (+34.7% vs +14.3%/yr) at realistic costs.
- The +1.5 Mkt- β is what the ML model decided was optimal – not a leverage knob we set. Alpha and β -tilt are joint outputs.
- β -hedged pure alpha (Sharpe +0.70) is small standalone but **uncorrelated with S&P** (corr ≈ 0) – a real diversifier in a multi-asset portfolio.

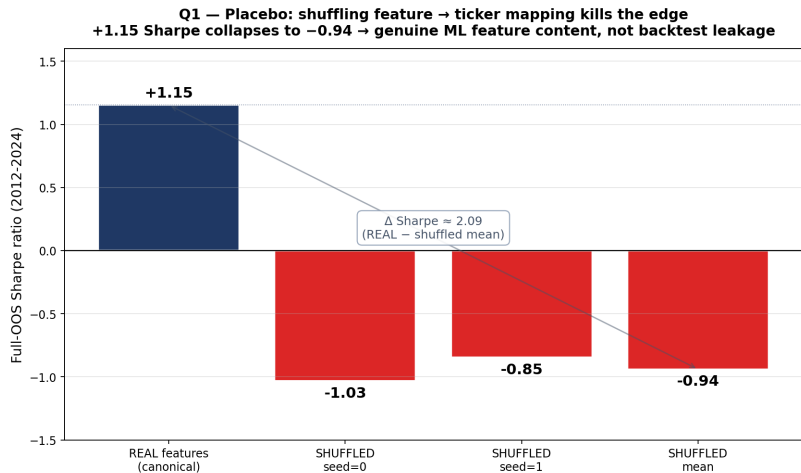
(N) Equity curve — three-line decomposition (read the RED line)



(N) Robustness battery — alpha survives every rigor check

Check	Result	Interpretation
Feature-shuffle placebo	+1.15 $\rightarrow$ -0.94	rules out engine / target / cost leakage
Carhart 6F (FF5+UMD)	α rises to +20.1%/yr at $t=+7.4$, UMD $\beta=-0.43$	momentum-AVERSE
Block bootstrap (6-mo blocks, 10k iter)	$P(SR \leq 0) = 0.0002$	excludes zero >99% confidence
Deflated Sharpe Ratio (N=25)	long-OOS	
Cost grid stress	DSR = 0.85-0.88	survives multiple-testing penalty
	α significant up to ~50 bps/side	survives conservative cost assumptions
Dense k-sweep + plateau zoom (CIs)	k=20 indistinguishable from every k in [10,20]	empirically defensible hyperparameter

(N) Placebo — the cleanest leakage test we have



Shuffling the feature → ticker mapping within each rebalance date kills the edge:
+1.15 Sharpe collapses to -0.94 (mean of 2 seeds). The strategy needs the real

(B) Regime overlay — works on strict-S&P, net-zero on broad book

- Andrea's workstream (she is not presenting today). **Gaussian HMM**, 2 states (calm/crisis), walk-forward expanding window with 60-mo burn-in, StandardScaler refit each step.
- **Overlay rule:** calm $\rightarrow 1.00\times$ leverage, crisis $\rightarrow 0.40\times$ (leverage-only).
- **Universe-dependent result:**
 - **Strict-S&P canonical:** DD improves $-25.5\% \rightarrow -19.9\%$ with small Sharpe gain $\square$
 - **Broad canonical:** DD unchanged at -33.8% $\square$

Why: the deepest drawdown is the Feb-Mar 2020 COVID crash; HMM flags March correctly but the overlay de-levers off the prior month-end label — January and February were both *calm*, so we entered the crash at full leverage. **Honest monthly-frequency timing limit, not a model flaw.**



(B) The audit story — Sharpe $+1.49 \rightarrow -0.31 \rightarrow +1.15$

- **Phase 14 (pre-audit):** S&P 500 union, $k=5$, apparent Sharpe $+1.49$. Looked great.
- **PIT audit week 4:** engine wasn't filtering by point-in-time S&P 500 membership. 726 non-member positions traded in a 2012–19 RandomModel run.
- **Phase 15 (PIT applied to S&P):** Sharpe -0.31 , α n.s. on every window. The $+1.49$ was 100% survivorship leak.
- **Rebuild on broad survivorship-free Sharadar universe:** Phase 23 $\rightarrow$ 24-RT, Sharpe $+1.15$, α significant at $t > 5$.
- **Two more bugs caught in week 5:**
 - Q-filter dropping NDAQ + IONQ (alive common stock) $\rightarrow$ corrected gate on `isdelisted=='Y'`
 - INCLUDE_FEATURES subset not applied $\rightarrow$ canonical was silently 16-feature $\rightarrow$ corrected to declared 14
- The **journey from $+1.49 \rightarrow +1.15$** is the methodological contribution.

(N) Honest counterweights — not market-neutral; alpha lives down-cap

- **Mkt- $\beta \approx +1.3$** — strategy is not market-neutral despite dollar-neutral construction. Long-leg makes +37.9%/yr; short-leg is a near-zero-P&L hedge (−2%/yr). **~55% of headline return is leveraged market exposure; ~45% is genuine cross-sectional skill (FF5-residual).**
- **Where the alpha lives** — strict rolling top-2,000 (large-cap end) gives $\alpha = +1.8\%/yr$ at $t=0.96$ (n.s.). SMB- β also collapses (+1.26 $\rightarrow$ +0.15). Confirms **GKX 2020 §IV.D**: ML alpha lives in the small/mid-cap tail.
- **Capacity binding at deployable AUM** — 175% turnover $\times$ ~440 small/mid-cap positions = ~770% NAV/mo throughput; realistic small-cap costs likely above 30 bps/side.
- **Single-name fragility** (LO study, $n=11$): top name (LSCG) moves Sharpe by +0.089. Suggestive, underpowered.

Q8 — Where the alpha lives: the headline +18.7%/yr is a DOWN-CAP effect (GKX 2020 §IV.D)
On the strict rolling top-2,000 alone, the FF5 alpha is +1.8%/yr at $t=0.96$ (not significant)

FF5 α (%/yr)

FF5 α t-stat

Mkt- β

SMB- β

(N) Conclusion + what we'd do differently

- **Defensible claim:** under realistic survivorship controls + PIT eligibility + 10 bps/side costs, the Phase 24-RT canonical produces statistically significant cross-sectional alpha on the 2012-2024 OOS sample, robust to **every** rigor check (Carhart, DSR, bootstrap, placebo, cost stress, k-sweep, sanity gates).
- **Honest bounds:** not market-neutral; capacity-constrained at deployable AUM; single-name fragility; regime overlay net-zero on broad book.
- **What we'd do differently** (in priority):
 - 1 **Hysteresis selection / turnover dampening** – the model re-picks top-k/bottom-k each month independently, kicking out marginal names (rank-19 last month, rank-22 this month). A buffer band (“keep last month’s longs as long as they’re still in top-30”) could cut turnover ~40% with <10% alpha loss, saving ~0.9 pp/yr at 10 bps and ~2.6 pp/yr at 30 bps.
 - 2 Size-impact-aware cost modelling (Almgren-Chriss) on the actual cap-bucket distribution
 - 3 Sub-monthly regime detection (weekly/daily HMM) to catch fast crashes
 - 4 Cliff-style leave-one-out on full universe ($n \gg 11$) with bootstrap CIs
- **What we learned:** the audit journey (+1.49 $\rightarrow$ -0.31 $\rightarrow$ +1.15) is the

Reference materials in the appendix:

- Full report `report/REPORT.md`
- Decision log `DECISIONS.md` (1,233 lines, complete chronological provenance)
- Q&A prep doc with 25 anticipated questions
- All robustness artefacts under `results/`

Repository: <https://github.com/bowenzuo119-hash/ml-factor-investing-2026>